Self-Preference Bias in LLM Juries: A Controlled Test of Hard Holdout

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Abstract

LLM-as-a-Judge ensembles can include the evaluated model in its own jury, creating a direct opportunity for self-preference bias. We test whether hard holdout*—excluding the generating model from its own jury at aggregation time—reduces this bias in subjective pairwise evaluation.*

Using 300 Chatbot Arena matchups and a seven-model jury spanning distinct providers, we collect 4,196 judge votes and compare inclusive and holdout verdicts using jury-level false endorsement rate (FER). We find no detectable FER benefit from hard holdout in this setting ($p = 0.875$, rank-biserial $r = 0.600$, $n_{\text{eff}} = 4$), despite clear vote-level self-preference in six of seven models under Wataoka's Equal Opportunity diagnostic.

This separation between vote-level bias and jury-level effect is the paper's main result. Secondary jury-size simulations and a simplified pivotal-voter model suggest that holdout is more relevant in very small panels, especially at $k = 3$, but we treat these analyses as explanatory rather than definitive. The practical implication is therefore narrow: for this diverse seven-model jury on subjective pairwise evaluation, hard holdout provides no detectable benefit, even though individual self-preference is real.

1 Introduction

The adoption of large language models as automated evaluators, LLM-as-a-Judge (LaaJ), has accelerated rapidly across benchmarking, reinforcement learning from human feedback, and alignment research. LaaJ reduces evaluation costs by orders of magnitude compared to human annotation while achieving high agreement with human preferences on standardized benchmarks [1]. However, this agreement masks systematic biases that can silently distort evaluation outcomes at scale.

Among these, **self-preference bias** is especially relevant for ensemble evaluation: LLMs may systematically overrate outputs from their own generation distribution. Wataoka et al. [2] quantify this effect using an Equal Opportunity metric from algorithmic fairness, showing that models assign disproportionately high scores to responses with lower perplexity, a proxy for text similar to their own outputs. The implication is that any evaluation pipeline in which the generating model also acts as a judge may inherit a built-in source of evaluator bias.

Ensemble judging has been proposed as a general mitigation for individual judge biases [3, 4]. By aggregating across multiple models, the variance and idiosyncratic biases of any single judge are reduced. However, **inclusive jury configurations**, where the generating model participates in judging its own outputs, preserve self-preference bias. The self-serving vote is simply averaged in alongside the votes of other judges.

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One possible intervention is to exclude the generating model from judging its own outputs. The closest work to this idea is Gu et al. [5], who propose entanglement-aware verifier reweighting: a soft, continuous downweighting of judges that are statistically dependent on the evaluated model. On MMLU-Pro, this approach reduces correlated evaluation error and improves ensemble accuracy by up to 4.5% over majority voting. Yet their intervention is (a) a soft reweighting rather than hard holdout, (b) evaluated only on objective multiple-choice tasks with known correct answers, and (c) measured with a precision-based bias metric rather than the fairness-grounded Equal Opportunity criterion.

This paper asks a closely related question: does the simpler intervention of *hard holdout* (programmatically removing the generating model from the judge panel entirely) reduce self-preference bias on subjective open-ended generation tasks, where ground truth is defined by human preference?

We answer this empirically. Using 300 pairwise matchups from the Chatbot Arena human preference dataset [6] and a pool of seven models spanning distinct training lineages, we collect all judge votes in a single inclusive run and derive the holdout condition mathematically by filtering out self-votes. Our primary outcome is jury-level false endorsement rate (FER), and our primary significance test compares inclusive and holdout FER across models using the Wilcoxon Signed-Rank Test [7]. Wataoka's Equal Opportunity metric is used separately as an individual-vote diagnostic. This distinction—vote-level self-preference versus jury-level effect—is the paper's central analytical contribution.

Our contributions are:

1. To our knowledge, the first controlled empirical test of hard holdout as a mitigation for self-preference bias in LLM-as-a-Judge ensembles on subjective open-ended evaluation.
2. A separation of jury-level and vote-level analysis: we test hard holdout on jury-level false endorsement while separately measuring self-preference with Wataoka's Equal Opportunity metric.
3. A curated dataset of 4,196 individual judge votes across seven model families, enabling alternative aggregation and bias analyses without repeating the API calls.

2 Related Work

2.1 LLM-as-a-Judge

The use of large language models as automated evaluators was systematized by Zheng et al. [1], who introduced MT-Bench and Chatbot Arena as benchmarks for open-ended chat assistant evaluation. They demonstrated that strong LLMs such as GPT-4 achieve over 80% agreement with human raters on pairwise preference judgements, establishing LLM-as-a-Judge (LaaJ) as a viable alternative to costly human annotation. Since then, LaaJ has been widely adopted across benchmarking pipelines, RLHF reward modeling, and automated red-teaming.

Despite its utility, LaaJ is subject to several systematic biases. Positional bias causes models to prefer responses in certain positions (e.g., Response A) independently of quality [1]. Verbosity bias leads models to favor longer responses regardless of content [1]. Self-preference bias, the focus of this work, causes models to systematically overrate outputs from their own generation distribution.

2.2 Self-Preference Bias

Wataoka et al. [2] provide the most rigorous quantification of self-preference bias to date. They propose a metric grounded in the Equal Opportunity criterion from algorithmic fairness:

Bias = P(Y′=1 $\mid S = 1, Y = 1)$
$- P(Y' = 1 \mid S = 0, Y = 1)$

where Y' is the judge's binary preference, S indicates whether the judge and generator are the same model, and Y is the human ground truth label. A bias of zero indicates fair evaluation; positive values indicate that the model inflates its own outputs relative to human judgement.

Wataoka et al. further identify perplexity as the underlying mechanism: LLMs assign lower perplexity to text that resembles their own generation distribution, and this familiarity systematically influences their evaluative judgements. This insight suggests that self-preference is not merely a surface artifact but a structural property of how language models process text.

2.3 Ensemble Judging

A natural mitigation for individual judge biases is to aggregate across multiple judges. Verga et al. [3] propose replacing single judges with inclusive juries, showing that ensemble aggregation improves agreement with human preferences over any single model. Chan et al. [4] similarly demonstrate that multi-agent debate-style evaluation yields more reliable assessments than single-judge setups.

However, these approaches generally treat aggregation as a variance-reduction mechanism, not as a targeted intervention against self-evaluation. When the generating model remains inside the panel, its self-serving vote still contributes to the verdict.

2.4 Entanglement-Aware Reweighting

The closest prior work to ours is Gu et al. [5], who identify a related but distinct problem: latent behavioral entanglement between models sharing training data, distillation pipelines, or alignment procedures. Entangled models tend to agree on incorrect answers, inflating apparent ensemble confidence without improving accuracy.

Gu et al. propose entanglement-aware verifier reweighting, which dynamically penalizes judges based on their pairwise entanglement score with the evaluated model. On MMLU-Pro, this approach achieves up to 4.5% accuracy improvement over standard majority voting. Crucially, their mechanism operates as a soft continuous downweighting rather than hard holdout, and is evaluated exclusively on objective multiple-choice tasks with known ground truth labels.

This leaves open the space our paper targets: a simpler intervention, hard holdout of the self-judge, evaluated on subjective open-ended comparisons rather than objective multiple-choice tasks.

3 Methodology

3.1 Experimental Design

We conduct a controlled experiment comparing self-preference bias under two jury configurations:

- **Inclusive jury (control):** All seven models in the pool judge every pairwise matchup, includ-

ing the model whose response appears in the matchup.
- **Hard holdout jury (treatment):** The generating model is excluded from judging matchups containing its own output. The final verdict is formed from the remaining six judges.

Both conditions are derived from a single jury run. We collect all seven judge votes per matchup, then compute the holdout condition by filtering out self-votes ($S = 1$) at the metric computation stage. This avoids running two separate API passes and ensures both conditions are evaluated on identical data.

3.2 Model Pool

We select seven models from distinct providers and model families that were competitive in the Chatbot Arena ecosystem during the study period. The goal is not to define a canonical "frontier" set, but to assemble a diverse evaluator pool rather than a cluster of near-duplicate judges.

This selection does not guarantee independence: large contemporary models inevitably share broad internet-scale training data and similar post-training conventions. The design goal is more modest. By spanning providers and model families, we aim to reduce the chance that the ensemble behaves like a tightly entangled single lineage.

3.3 Data

We use the **lmarena-ai/arena-human-preference-140k** dataset [6], a publicly available collection of 135,634 pairwise human preference judgements collected from Chatbot Arena. Each row represents a single human vote comparing two model responses on a user-provided prompt.

We filter to matchups where both `model_a` and `model_b` are members of our model pool (`evaluation_order == 1` to ensure single-turn prompts; `winner` $\in$ {model_a, model_b, tie} to exclude ambiguous `both_bad` rows). This yields 3,342 eligible matchups. We sample 300 uniformly, providing sufficient statistical power while remaining within budget.

Human preference votes serve as the ground truth quality label Y in Wataoka's Equal Opportunity metric.

3.4 Judging Protocol

Each judge receives a standardized prompt consisting of a system instruction, the user's original prompt, Response A, and Response B. Judges are instructed to reply with a single word: **A**, **B**, or **tie**.

```
System: You are an impartial judge
evaluating the quality of two AI
assistant responses. Given a user
prompt and two responses (Response A
and Response B), decide which is better.
Reply with exactly one word: A, B, or
tie.
```

Temperature is set to 0 for determinism. All calls are routed through the OpenRouter API to ensure consistent infrastructure across providers.

Position-swapping: Each matchup is presented twice, as (A, B) and as (B, A). Votes from swapped presentations are normalized back into the original A/B orientation at collection time. For the analyses reported here, metric computation uses the original-orientation vote set, while the swapped pass serves as a positional-bias control and audit trail [1].

Judges may also answer **tie**. Throughout the paper, ties are treated as abstentions from the A/B majority count. This matters substantively because high tie rates reduce the effective number of decisive votes in a jury and help explain some of the model-level and jury-size patterns discussed later.

Each model's 14 calls per matchup (7 judges $\times$ 2 positions) are dispatched concurrently.

3.5 Bias Metrics

Our primary metric for the main hypothesis is **jury-level false endorsement rate (FER)**:

$$\mathrm{FER}_m = P(\hat{y}_{jury} = m \mid y_{human} \neq m) \tag{1}$$

For model m, FER is computed across all non-tie matchups in which m appears and the human preferred the competing response. Lower FER indicates less bias in the false-endorsement direction. We compare FER under two conditions:

- **Inclusive FER:** all eligible judges contribute, including the self-judge.
- **Holdout FER:** the self-judge is excluded before forming the jury verdict.

As a separate vote-level diagnostic, we adopt Wataoka et al.'s [2] Equal Opportunity self-preference metric:

$\mathrm{Bias}_m = P(Y' = 1 \mid S = 1, Y = 1)$
$- P(Y' = 1 \mid S = 0, Y = 1)$

where for model m: $Y' = 1$ if the judge's preference matches m's response; $S = 1$ if the judge and generator are the same model; $Y = 1$ if the human voter also preferred m's response. We use this diagnostic to evaluate whether self-preference exists at the individual-vote level, not as the primary test of holdout effectiveness.

Table 1: *Model pool with provider and API identifiers.*

Model	Provider	Arena ID
GPT-4.1	OpenAI	`gpt-4.1-2025-04-14`
Claude Opus 4	Anthropic	`claude-opus-4-20250514`
Gemini 2.5 Pro	Google	`gemini-2.5-pro`
Grok 3	xAI	`grok-3-preview-02-24`
DeepSeek V3	DeepSeek	`deepseek-v3-0324`
Llama 4 Maverick	Meta	`llama-4-maverick-03-26-experimental`
Mistral Medium	Mistral	`mistral-medium-2505`

3.6 Statistical Testing

For the primary hypothesis, we compare per-model FER scores between inclusive and holdout conditions using the **Wilcoxon Signed-Rank Test** [7], a non-parametric paired test appropriate for the small sample of models ($n = 7$). The alternative hypothesis is one-sided:

$$H_1 : \text{FER}_{inclusive} > \text{FER}_{holdout} \tag{2}$$

That is, hard holdout reduces false endorsement.

For the secondary hypothesis H_2 (vote-level self-preference is positive across models), we examine Wataoka bias descriptively rather than through a separate significance test.

Effect size for the Wilcoxon test is reported as rank-biserial correlation r. We use a significance threshold of $\alpha = 0.05$.

4 Results

4.1 Primary Result: Jury-Level False Endorsement

Table 2 reports **False Endorsement Rate (FER)** for each model under both the inclusive (self-judge included) and holdout (self-judge excluded) conditions.

Table 2: *Jury-level False Endorsement Rate (FER) under inclusive and holdout conditions.*

Model	Inc. FER	Hol. FER	Δ
Claude Opus 4	0.264	0.278	−0.014
DeepSeek V3	0.188	0.250	−0.062
Gemini 2.5 Pro	0.731	0.731	0.000
GPT-4.1	0.122	0.098	+0.024
Grok 3	0.286	0.333	−0.048
Llama 4 Maverick	0.517	0.517	0.000
Mistral Medium	0.211	0.211	0.000

Three models (Gemini, Llama, Mistral) show zero change. Three models (Claude, DeepSeek, Grok) show increased FER under holdout; holdout is slightly worse. Only GPT-4.1 shows a marginal improvement (+0.024). There is no consistent direction. The three zero-difference cases arise because removing the self-vote never changed the strict-majority verdict in the relevant FER subset, either because peer votes already determined the outcome or because the self-vote aligned with the eventual majority.

A Wilcoxon Signed-Rank Test (one-sided; alternative: inclusive FER > holdout FER) yields $W = 2.0$, $p = 0.875$, $r = 0.600$ ($n_{\text{eff}} = 4$ after excluding the three zero-difference pairs). **H1 is not supported.** In this seven-model setting, hard holdout yields no detectable reduction in jury-level false endorsement.

The large effect size ($r = 0.600$) is reconciled by the low statistical power of the test: with only $n_{\text{eff}} = 4$ non-zero pairs, the Wilcoxon test cannot achieve $p < 0.05$ even for moderate effects. We therefore interpret this as a no-detectable-benefit result in the current setup, not as strong evidence that hard holdout is universally ineffective.

4.2 Individual-Vote Self-Preference

Although the jury-level test is null, self-preference is detectable at the individual vote level. We apply Wataoka et al.'s [2] Equal Opportunity metric (Equation 2). Positive bias means that, conditional on the human preferring m, the self-judge endorses m more often than peer judges do.

Six of seven models show positive individual-vote self-preference (**H2 supported**). Claude (+0.272) and Gemini (+0.254) are the strongest: their self-votes agree with human preference at a rate roughly 25–27 percentage points higher than peer judges do, conditional on the human preferring their output.

Mistral Medium is a consistent anomaly: a large negative bias (−0.350), meaning Mistral self-votes are *less* likely to agree with human preference than peer votes. This appears closely tied to its tendency to vote "tie" in ambiguous cases; in the Wataoka metric, those tie votes count as disagreement with the human winner.

4.3 Diagnostic Metrics

Three additional metrics provide complementary views of model-level self-preference.

Table 3: *Vote-level self-preference (Wataoka Equal Opportunity metric). P_s = self-judge agreement rate; P_o = peer-judge agreement rate, both conditional on the human preferring model m.*

Model	P_s	P_o	Bias	N
Claude Opus 4	0.667	0.394	$+$**0.272**	30
DeepSeek V3	0.550	0.425	$+0.125$	20
Gemini 2.5 Pro	0.922	0.668	$+$**0.254**	64
GPT-4.1	0.417	0.396	$+0.021$	24
Grok 3	0.610	0.541	$+0.069$	41
Llama 4 Maverick	0.720	0.680	$+0.040$	25
Mistral Medium	0.103	0.453	$-$**0.350**	39

Jury Accuracy vs. Human Preference We measure the rate at which the jury majority matches the human winner:

$$\mathrm{Acc}_m = P(\hat{y}_{jury} = y_{human} \mid m \in matchup) \tag{3}$$

Table 4: *Jury accuracy under inclusive and holdout conditions.*

Model	Inc.	Hol.	Δ	N
Claude Opus 4	0.500	0.516	$+0.016$	124
DeepSeek V3	0.426	0.407	-0.019	54
Gemini 2.5 Pro	0.513	0.549	$+0.035$	113
GPT-4.1	0.571	0.623	$+$**0.052**	77
Grok 3	0.571	0.614	$+0.043$	70
Llama 4 Maverick	0.500	0.531	$+0.031$	64
Mistral Medium	0.459	0.459	0.000	98

Five of seven models show improved accuracy under holdout. GPT-4.1 ($+0.052$) and Grok ($+0.043$) show the largest improvements. We treat this as descriptive rather than decisive: accuracy is not the primary endpoint of this study.

Self-Vote Deviation Self-vote deviation measures the rate at which the self-judge uniquely votes for its own model when the rest of the panel disagrees:

$$\mathrm{Dev}_m = P(self \to m \ \land \ peers \not\to m \mid m \in matchup) \tag{4}$$

Table 5: *Self-vote deviation (self-favoring votes against panel majority).*

Model	Self-favor rate	N
Claude Opus 4	**0.347**	124
Llama 4 Maverick	0.188	64
DeepSeek V3	0.185	54
Gemini 2.5 Pro	0.124	113
GPT-4.1	0.117	77
Mistral Medium	0.061	98
Grok 3	0.057	70

Claude is a strong outlier: in 34.7% of matchups where it is the evaluated model, it votes for itself against the panel majority. This behavioral signature is consistent with its high Wataoka bias ($+0.272$).

Score Inflation Score inflation measures the mean difference between the self-judge's endorsement rate and the peer-panel's endorsement rate across all matchups where model m appears.

Table 6: *Score inflation: self-judge endorsement rate minus peer-panel endorsement rate.*

Model	Mean inflation	N
Claude Opus 4	$+$**0.298**	124
Gemini 2.5 Pro	$+$**0.263**	113
DeepSeek V3	$+0.151$	54
Llama 4 Maverick	$+0.112$	64
Grok 3	$+0.036$	70
GPT-4.1	0.000	77
Mistral Medium	$-$**0.291**	98

Claude ($+0.298$) and Gemini ($+0.263$) rate their own outputs nearly 0.30 higher than their peer panel does on average. GPT-4.1 shows exactly zero inflation, a genuinely neutral self-judge. Across all four metrics, Claude and Gemini are consistently the highest self-preference models; Mistral is consistently the anomaly; GPT-4.1 is consistently neutral.

4.4 Secondary Jury-Size Analysis

The primary experiment fixes the jury size at $n = 7$. As a secondary explanatory analysis, we simulate FER at $k = 2, \ldots, 7$ using the observed votes to see how the *potential* holdout benefit changes with jury size inside this dataset. For each k, the inclusive jury always contains the self-judge plus $k - 1$ peers sampled without replacement from the remaining six; the holdout jury contains only those $k - 1$ peers. We run 1,000 iterations per k with a fixed seed.

The clearest small-jury result is at $k = 3$ ($\Delta = +0.093$, 95% CI excludes zero by a wide margin). At $k = 3$, the self-judge constitutes one-third of the inclusive vote and can be decisive when the two peers split 1-vs-1. The result at $k = 6$ ($\Delta = +0.043$,

Table 7: *Jury-size simulation: FER under inclusive and holdout configurations (1,000 iterations per k). At $k = 7$ the result is fixed (no resampling).*

k	Inc. FER	Hol. FER	Δ	95% CI
2	0.299	0.325	-0.026	$[-0.062, +0.008]$
3	0.349	0.256	$+\mathbf{0.093}$	$[+\mathbf{0.062}, +\mathbf{0.128}]$
4	0.321	0.299	$+0.022$	$[-0.008, +0.054]$
5	0.312	0.286	$+0.026$	$[+0.000, +0.049]$
6	0.328	0.285	$+\mathbf{0.043}$	$[+\mathbf{0.021}, +\mathbf{0.066}]$
7	0.309	0.317	-0.008	[fixed]

CI $[+0.021, +0.066]$) is suggestive but should be interpreted cautiously. The simulation at $k = 7$ ($\Delta = -0.008$) reproduces the primary result exactly, but this simulation should be read as secondary context for the main experiment rather than as the main basis for the paper's recommendation.

4.5 Heuristic Theoretical Analysis

To provide intuition for the jury-size pattern, we derive a simplified formula for the holdout benefit $\Delta(n)$ under a binary IID vote model. Each judge independently endorses the rejected model M with probability p_s (self-judge) or p_p (each peer, IID). The jury uses strict majority; tie votes abstain from the A/B count.

Pivotal-voter formula (odd n):

$$\Delta(n) = p_s \cdot n - 1(n-1)/2 \cdot \left[p_p(1-p_p)\right]^{(n-1)/2} \quad (5)$$

The holdout benefit equals the self-preference rate p_s multiplied by the probability that peer judges deadlock at exactly $(n-1)/2$ votes each, the only scenario in which the self-judge is the decisive voter. Because the observed data include heterogeneous peers and substantial tie rates, we treat this model as a heuristic explanation rather than a strong predictive account.

We estimate $p_s = 0.444$ and $p_p = 0.325$ from the rejection-condition votes. The theoretical Δ values for selected odd jury sizes are shown in Table 8.

Table 8: *Theoretical holdout benefit $\Delta(n)$ under the IID pivotal-voter model.*

n	FER_{inc}	FER_{hol}	$\Delta(n)$
3	0.300	0.105	$+\mathbf{0.195}$
5	0.232	0.104	$+0.128$
7	0.186	0.092	$+0.094$
13	0.104	0.058	$+0.046$
23	0.044	0.027	$+0.018$
31	0.023	0.014	$+0.009$

The theoretical model predicts that $\Delta(n)$ falls below 0.05 at $n = 13$ and below 0.01 at $n = 31$. Despite a systematic quantitative gap (the IID model overestimates the simulation by approximately 0.10), the model and simulation point in the same qualitative direction: the holdout benefit should shrink as jury size grows. We treat that directional agreement, rather than the numerical thresholds themselves, as the main value of the theory section.

4.6 Summary of Findings

The results support three takeaways. First, self-preference is clearly present at the individual-vote level: six of seven models show positive Wataoka bias, with Claude and Gemini standing out as the strongest cases. Second, that vote-level bias does not translate into a measurable jury-level FER improvement from hard holdout at $n = 7$. Third, the secondary jury-size analyses suggest that holdout is most relevant in very small panels, especially at $k = 3$, while evidence at larger small-jury sizes is weaker and more assumption-sensitive.

5 Discussion

5.1 Why Holdout Does Not Help at n=7

The null result at $n = 7$ becomes more interpretable in light of the theory. The pivotal-voter formula (Section 4.5) gives a theoretical $\Delta(7) = 0.094$ under IID assumptions, but the empirical simulation yields $\Delta = -0.008$. The gap is explained by two structural features of the data.

First, peer votes are heterogeneous. The IID model assumes all peer judges endorse the rejected model with the same probability $p_p = 0.325$. In reality, peer endorsement rates vary substantially across models, so the deadlock probability is not governed by a single shared parameter.

Second, high tie rates reduce the effective jury size. Mistral Medium votes “tie” in a large fraction of cases, effectively abstaining from the A/B count. When several judges abstain, the nominal jury size of seven collapses to an effective size of four or five, changing the deadlock arithmetic and introducing the even-n threshold asymmetry at unexpected points.

Together, these two factors push the empirical $\Delta(7)$ well below the IID theoretical prediction. The self-preference bias does not disappear; it remains detectable at the vote level, but in this specific seven-model setting the jury's aggregation over six heterogeneous peers is sufficient to absorb it.

5.2 When Holdout Does Help

The simulation at $k = 3$ ($\Delta = +0.093$) identifies the clearest regime where hard holdout is effective: very small juries where the self-judge constitutes a substantial fraction of the decisive vote. At $k = 3$, the self-judge is one-third of the panel and is pivotal whenever the two peers split 1-vs-1. Removing the self-judge directly eliminates that decisive vote.

This has a practical implication for evaluation design. When budget or latency constraints force very small judge panels, hard holdout is a low-cost, zero-assumption intervention worth considering. The evidence is strongest for three-judge panels; for larger small-jury settings, the current study is suggestive rather than definitive.

The boundary between "holdout helps" and "holdout does not help" is not a fixed jury size but a function of the self-preference rate (p_s) and peer endorsement rate (p_p) of the specific models involved. The pivotal-voter formula therefore gives a model-specific way to reason about when holdout is likely to become negligible.

5.3 Comparison with Gu et al.

Our results complement rather than contradict Gu et al. [5]. The two studies ask related questions under importantly different conditions.

Task type. Gu et al. evaluate on MMLU-Pro, an objective multiple-choice benchmark with known ground truth. Our task is open-ended pairwise comparison where ground truth is human preference, a noisier and more subjective signal. That alone makes it harder for any bias-mitigation intervention to generate large, clean gains at the jury level.

Intervention type. Soft entanglement-aware reweighting continuously penalizes correlated judges without removing any votes. Hard exclusion removes the self-judge's vote entirely. The interventions are aimed at different dependence structures: Gu et al. address graded entanglement, while our setting isolates the special case of exact judge-generator identity.

Panel structure. Our panel deliberately spans multiple providers and families. By contrast, our null result at $n = 7$ is consistent with a panel where diversity already weakens any one model's influence on the aggregate verdict.

The broader lesson is that dependence-aware evaluation is task- and panel-dependent. Gu et al.'s method is better suited to graded entanglement on objective tasks; hard holdout is most plausible when self-judging is the main concern and the jury is small enough for one vote to matter.

5.4 Model-Level Variation

A secondary finding that warrants attention is the substantial heterogeneity in self-preference across models. Claude Opus 4 and Gemini 2.5 Pro show consistently high self-preference across all four metrics: high Wataoka bias (+0.27, +0.25), high score inflation (+0.30, +0.26), and elevated self-vote deviation (34.7%, 12.4%). GPT-4.1 is the opposite extreme: negligible Wataoka bias (+0.02), zero inflation, and low deviation (11.7%). Mistral Medium is anomalous in the anti-self-preference direction across all metrics.

This heterogeneity matters for evaluation design even apart from the main holdout question. A single-judge or very small-jury setup is much more exposed to the choice of evaluator than a seven-model ensemble.

5.5 Limitations

Sample size. The Wilcoxon test operates on $n = 7$ per-model observations (reduced to $n_{\text{eff}} = 4$ after zero-difference pairs). Power to detect effects below $r \approx 0.8$ is low. A null result with $r = 0.600$ is therefore consistent with both "truly no effect" and "a small-to-moderate effect that this study is underpowered to detect reliably."

Model pool as of early 2026. The seven models represent a particular generation of strong contemporary LLMs. Self-preference rates may differ substantially for smaller models, instruction-tuned variants, or future model families.

Single API run. Determinism ($T = 0$) reduces variance, but a single run means we cannot estimate within-model variance in judge decisions.

Simulation generalizability. The jury-size simulation resamples from existing votes rather than recruiting purpose-built smaller panels. A three-judge panel designed from the ground up may exhibit different FER characteristics.

Position swapping. We mitigate positional bias by collecting both (A, B) and (B, A) presentations. The analyses reported here use the original-orientation vote set, leaving open the possibility that averaging both passes could change some secondary estimates.

6 Conclusion

We investigated whether hard holdout (programmatically excluding the evaluated model from its own jury) reduces self-preference bias in LLM-as-a-Judge ensembles. In a diverse seven-model jury on subjective pairwise evaluation, we find no detectable jury-level FER benefit from hard holdout ($p = 0.875$, $r = 0.600$, $n_{\text{eff}} = 4$), despite clear vote-level self-preference in six of seven models.

That separation between vote-level bias and jury-level effect is the paper's main contribution. Individual models can display substantial self-preference, yet aggregation across six peer judges can still absorb that bias well enough that hard holdout does not produce a detectable FER improvement in this setting.

The secondary simulation and heuristic theory suggest that the picture may differ in very small panels, especially at $k = 3$, where the self-judge is more often decisive. We therefore treat the practical implication as narrow rather than general: for this diverse seven-model jury on subjective pairwise evaluation, hard holdout shows no detectable benefit; in very small juries, it remains a plausible robustness intervention.

The limitations are direct. The primary test is based on only seven models, ties materially affect effective jury size, and the jury-size analyses are secondary rather than standalone experiments. Future work should test hard holdout prospectively with purpose-built small juries, replicate across additional task domains, and examine whether the strong vote-level self-preference observed for Claude and Gemini persists across model versions and prompting strategies.

Data and Code Availability

The vote dataset and sampled matchups are available at `huggingface.co/datasets/clarsam/llm-judge-holdout`. The code used to run the experiment and reproduce the analyses is available at `github.com/Walsamer/llm-judge-holdout`.

References

[1] Lianmin Zheng et al. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena". In: *arXiv preprint arXiv:2306.05685* (2023).

[2] Koki Wataoka et al. "Self-Preference Bias in LLM-as-a-Judge". In: *arXiv preprint arXiv:2410.21819* (2024). NeurIPS 2024 Safe Generative AI Workshop.

[3] Pat Verga et al. "Replacing Judges with Juries: Evaluating LLM Generations with a Panel of Diverse Models". In: *arXiv preprint arXiv:2404.18796* (2024).

[4] Chi-Min Chan et al. "ChatEval: Towards Better LLM-Based Evaluators through Multi-Agent Debate". In: *arXiv preprint arXiv:2308.07201* (2023).

[5] Hanze Gu et al. "Entanglement-Aware LLM Ensemble Reweighting". In: *arXiv preprint arXiv:2604.07650* (2026).

[6] LMSYS Chatbot Arena Team. *lmarena-ai/arena-human-preference-140k*. 2025. URL: `https://huggingface.co/datasets/lmarena-ai/arena-human-preference-140k`.

[7] Frank Wilcoxon. "Individual Comparisons by Ranking Methods". In: *Biometrics Bulletin* 1.6 (1945), pp. 80–83.